

Problems: Defects in crystalline structures

1. The fraction of vacancies in an aluminum lattice is $2.29 \cdot 10^{-5}$ at 400°C . Calculate the fraction of vacancies at 660°C (just before the melting point). $R = 8.31 \text{ J/mol}\cdot\text{K}$.
(Solution: 4.53×10^{-4})
2. Considering that the energies for Schottky and Frenkel defects formation in a specific material are 1 and 4 eV per atom, respectively. Determine the equilibrium concentrations of such defects at 1000K. $K = 8.62 \times 10^{-5} \text{ eV/K}$.
(Solution: 3.026×10^{-3} and 8.386×10^{-11})
3. Calculate the number of Mg^{2+} vacancies produced by the dissolution of:
 - a) 1 mole of Al_2O_3 in 99 moles of MgO. **(Solution: 0.01 mol vacancies/mol MgO)**
 - b) 2 moles of TiO_2 in 98 moles of MgO. **(Solution: 0.02 mol vacancies/mol MgO)**
4. Using data in the table below, order the elements according to their solubility in iron, from higher to lower solubility. Justify your answer. **(Solution: $\text{Mo} > \text{Ni} > \text{Mn}$)**

Element	Atomic radius (nm)	Crystalline Structure	Electronegativity	Valence
Iron	0.124	BCC	1.7	+2, +3
Nickel	0.125	FCC	1.8	+2
Molybdenum	0.136	BCC	1.3	+3, +4, +6
Manganese	0.112	Simple cubic	1.6	+2, +3, +6, +7

5. In a specimen of iron oxide (II), FeO , 10% of the Fe^{2+} ions are substituted by Fe^{3+} ions. In order to maintain the crystal electro neutrality, a fraction of the cationic positions are vacancies. Data: $M(\text{O})=16$; $M(\text{Fe})=55.8$.
 - a) Calculate the fraction of cationic vacancies. **(Solution: 0.033 mol vacancies/mol Fe^{2+})**
 - b) Assuming that the lattice parameter of the perfect and imperfect crystal is the same, calculate how much the density decreases, in %, with respect to the perfect crystal. **(Solution: -2.56%)**
6. Calculate the Burgers vector magnitude for $\alpha\text{-Fe}$ (BCC) and Al (FCC). **(Solution: $\alpha\text{-Fe}: \sqrt{3}a/2$; Al: $a/\sqrt{2}$)**
7. For a metal with FCC structure, E_1 and E_2 are the energies necessary to generate dislocations in the directions $[100]$ and $[110]$, respectively. Demonstrate that $E_1/E_2 = 2$. **(Solution: theory)**
8. A metallic crystal with FCC structure and lattice parameter $3.6 \cdot 10^{-10} \text{ m}$ contains an edge dislocation in the plane (110) . Determine the magnitude and the direction of the Burgers vector.
(Solution: $B = 2.553 \times 10^{-10}$, $[110]$).
9.
 - a) Determine the interplanar spacing and the length of the Burgers vector for slipping in the slip system $(110)/[\bar{1}\bar{1}1]$ in BCC tantalum.
 - b) Repeat assuming, that the slip system is $(111)/[1\bar{1}0]$.
Data: $a = 3.3026 \text{ \AA}$. **(Solution: a) $B = 2.860 \text{ \AA}$, $d_{(110)} = 2.335 \text{ \AA}$; b) $B = 4.671 \text{ \AA}$, $d_{(110)} = 1.907 \text{ \AA}$)**
10. Starting from a perfect MgO crystal, a solid solution of Al_2O_3 and MgO is prepared with an atomic proportion of 15:85 of Al:Mg. Calculate:
 - a) The number of vacancies for each Mg atom. **(Solution: 0.088 vacancies /atoms Mg)**
 - b) The % variation of density. **(Solution: -3%)**

Data: atomic masses: Al=26.98; Mg=24.31; O=16.00

11.- Calculate the number of vacancies per cm^3 expected in copper at 1080°C (just below the melting temperature). The energy for vacancy formation is 20000 cal/mol . **(Solution: $4.98 \times 10^{19} \text{ vacancies/cm}^3$)**

Data: Copper has an FCC crystal structure and a lattice parameter of $3.6151 \text{ \AA}$; $R = 1.987 \text{ cal/Kmol}$.

12.- The density of a sample of FCC palladium is 11.98 g/cm^3 , and its lattice parameter is $3.8902 \text{ \AA}$. Calculate (a) the fraction of the lattice points that contain vacancies and (b) the total number of vacancies in a cubic centimeter of Pd. **(Solution: a) 0.00204 b) $1.39 \times 10^{20} \text{ vacancies/cm}^3$)**

Data: $M_{\text{Pd}} = 106.4 \text{ g/mol}$; $N_A = 6.022 \times 10^{23} \text{ atoms/mol}$.